

Calculus Homework – Real series

Exercise 1. Show that the following real series are divergent!

$$(1) \sum_{n=1}^{\infty} \frac{n^2}{n+1}, \quad (2) \sum_{n=1}^{\infty} (-2)^{n+1}, \quad (3) \sum_{n=1}^{\infty} \sqrt{n^2 + 2n} - \sqrt{n}, \quad (4) \sum_{n=1}^{\infty} \frac{\log(n)}{n}$$

Exercise 2. Construct a family $\{[a_n, b_n] \mid a_n < b_n, a_n, b_n \in \mathbb{R}, n \in \mathbb{N}\}$ of closed real intervals such that

$$\mathbb{Q} \subseteq \bigcup_{n=1}^{\infty} [a_n, b_n] \quad \text{and} \quad \sum_{n=1}^{\infty} (b_n - a_n) < 1.$$

Exercise 3. Let $k \in \mathbb{N}$ and $q \in \mathbb{R}$ be fixed such that $|q| < 1$. Show that

$$\sum_{n=k}^{\infty} q^n = \frac{q^k}{1-q}$$

holds!

Exercise 4. Investigate the following real series for convergence! Calculate the sum of geometric and exponential series!

$$\begin{aligned} (4.1) \sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n}, \quad (4.2) \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}, \quad (4.3) \sum_{n=1}^{\infty} \frac{1}{n^2 + 3n}, \quad (4.4) \sum_{n=1}^{\infty} \frac{(-2)^{n+1}}{(n+1)^n} \\ (4.5) \sum_{n=2}^{\infty} \frac{1}{n \log(n)}, \quad (4.6) \sum_{n=3}^{\infty} \frac{(-1)^n}{5^{n-2}}, \quad (4.7) \sum_{n=1}^{\infty} (-1)^n (\sqrt{n+1} - \sqrt{n}) \\ (4.8) \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}, \quad (4.9) \sum_{n=0}^{\infty} \frac{3^{n-2}}{n!}, \quad (4.10) \sum_{n=1}^{\infty} \frac{n^3 + 1}{4n^4 + n^2}, \quad (4.11) \sum_{n=1}^{\infty} \frac{1}{n(n+2)} \\ (4.12) \sum_{n=1}^{\infty} \frac{2 \cdot (-3)^n}{4^{2n+1}}, \quad (4.13) \sum_{n=0}^{\infty} \frac{\sqrt{2n^2 + 1}}{n+3}, \quad (4.14) \sum_{n=1}^{\infty} \frac{n^2 + 4n + 1}{2^{3n+2}}, \quad (4.15) \sum_{n=1}^{\infty} \frac{2^n n!}{n^n} \\ (4.16) \sum_{n=1}^{\infty} \binom{2n}{n} \frac{1}{2^n}, \quad (4.17) \sum_{n=0}^{\infty} \left(\frac{n-1}{n+1} \right)^{n^2}, \quad (4.18) \sum_{n=1}^{\infty} \frac{n!}{n^n}, \quad (4.19) \sum_{n=1}^{\infty} \frac{(n+1)^2}{3n^4 + n} \\ (4.20) \sum_{n=2}^{\infty} \frac{1}{n^2 \log(n)}, \quad (4.21) \sum_{n=1}^{\infty} \frac{n^2 2^n - 6n!}{n^2 n!}, \quad (4.22) \sum_{n=1}^{\infty} \frac{n^{17}}{2^n + 3^n}, \quad (4.23) \sum_{n=4}^{\infty} \frac{3^{n-3}}{(-2)^{2n}} \\ (4.24) \sum_{n=1}^{\infty} \log \left(\frac{n}{n+1} \right), \quad (4.25) \sum_{n=2}^{\infty} \frac{\log(1 + \frac{1}{n})}{\log(n) \log(n+1)}, \quad (4.26) \sum_{n=1}^{\infty} \frac{2^n + 3^n}{6^n} \end{aligned}$$

Exercise 5. For which values of the real parameter p will the following real series be convergent?

$$\sum_{n=1}^{\infty} \left(\frac{p^2 + 2p + 1}{p + 2} \right)^n, \quad \sum_{n=2}^{\infty} \frac{1}{(\log(n))^p \cdot n}$$